

Reading Worksheet — Concrete Problems in AI Safety

Amodei, Olah, Steinhardt, Christiano, Schulman & Mané (2016), arXiv:1606.06565 — companion to the class reading guide

Name _____ Date _____

Fill this in by hand while the PDF is open on your device — write, don't type. Sketchy, incomplete answers are fine; the point is to have something to compare when we discuss it in class.

Pass 1 — Read the Bookends (~5 min: title, abstract, skim the conclusion)

The abstract's last sentence makes a promise without answering it. What is that promise? (*paraphrase is fine*)

Does the conclusion actually deliver on that promise? What does it say?

Your one-sentence claim for the whole paper:

Pass 2 — Read the Map (~10 min: intro + every section heading + figure/table captions)

List the section headings, in order, as they appear:

What organizing principle does Section 2 set up — the thing the five problem sections all map onto?

Sort the five problems under it:

Wrong objective function →

Objective function too expensive to evaluate →

Bad behavior during learning →

Pick one problem section (not your Pass 3 pick) and describe its shape in 2–3 phrases — example → failure → generalization → approaches → ?

Consolidate — Define These in Your Own Words

objective function

side effects

reward hacking

scalable oversight

safe exploration

distributional shift

Pass 3 — Go Deep (pick your own depth)

Which section are you going deep on, and why?

Optional LLM check — fill in the blanks, ask it, then verify against the actual text:

"I'm reading Amodei et al. 2016 and thinking about _____. In my own words, this means _____, and an analogy would be _____. Am I on the right path?"

Where did the LLM's answer differ from what the section actually says?

In your own words: what is an impact regularizer?

"Look up later" list — terms or citations you skipped past on purpose:

Bring to class

- Your one-sentence claim for the paper
- The five problems and their three-way grouping
- The section you went deep on, and why